

Strategic Risk Management and Long-Term Positioning: Planning Under Deep Uncertainty

Executive Summary

Strategic planning under uncertainty is the organization's most demanding cognitive task. The competitive future is fundamentally unknowable — yet organizations must commit resources today based on beliefs about tomorrow. Under Active Inference, strategic planning is **multi-model evaluation** over long time horizons, selecting strategies that perform well across a range of possible futures. This module covers scenario planning, strategic risk management, real options, and the art of building resilient strategies.

Learning Objectives

1. Apply **scenario planning** to evaluate strategies across multiple possible futures
 2. Understand **real options** thinking — preserving strategic flexibility
 3. Design **risk management** systems for strategic-level risks
 4. Build **resilient strategies** that perform adequately across a wide range of futures
 5. Balance **long-term vision** with **short-term adaptability**
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Key Concepts

1. Scenario Planning for Strategy

Strategic scenarios are not predictions — they are **alternative models of how the future might unfold**, used to stress-test strategies:

Step	Process
1. Identify key uncertainties	What market/technology/regulatory factors are most uncertain?
2. Construct scenarios	Build 3-4 internally consistent future narratives
3. Evaluate strategy under each	Which strategy performs best across scenarios?
4. Identify robust elements	What strategy components work in all scenarios?
5. Define trigger signposts	What early indicators would tell us which scenario is unfolding?

2. Real Options Thinking

A real option is the right — but not the obligation — to take a strategic action in the future.
Strategic options thinking:

- **Values flexibility:** An option to enter a market later is worth something now
- **Stages investment:** Invest in stages, with decision points at each stage
- **Preserves upside:** Maintains access to positive outcomes without committing to downside risk

Case Study — Intel's EUV Investment: Intel invested billions in Extreme Ultraviolet lithography research not because the technology was ready, but to preserve the option to use it when it matured. This real option — costly to maintain but strategically invaluable — ensured Intel could match competitors when the technology became production-ready.

3. Strategic Risk Management

Risk Level	Time Horizon	Examples	Management Approach
Operational	Weeks-months	Supply disruption, quality issues	Standard risk management
Strategic	Years	Technology disruption, market shift	Scenario planning, options
Existential	Decades	Industry obsolescence, regulatory change	Strategic transformation

4. Resilient vs. Fragile Strategies

Dimension	Fragile Strategy	Resilient Strategy
Assumptions	Requires specific future to unfold	Performs adequately across multiple futures
Flexibility	Heavily committed, hard to change course	Maintains options and adaptability
Information	Ignores disconfirming signals	Actively seeks and responds to new information
Portfolio	Concentrated bet on one scenario	Diversified across strategic options

Cross-References

- For organizational planning, see [Organizational Systems: Planning](#)
 - For collective foresight, see [Collective Intelligence: Planning](#)
 - For digital transformation roadmaps, see [Digital Transformation: Planning](#)
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Summary

Concept	Strategic Modeling Meaning
Scenario planning	Evaluating strategy across multiple possible futures
Real options	Preserving strategic flexibility by staging investment
Strategic risk	Risks to the viability of the organization's competitive model
Resilient strategy	Strategy that performs adequately across a range of futures
Signpost monitoring	Tracking indicators to determine which future is unfolding

References

- Schwartz, P. (1991). *The Art of the Long View*. Doubleday.
- Dixit, A. K., & Pindyck, R. S. (1994). *Investment Under Uncertainty*. Princeton University Press.